# Final Report: Arel Global Market Place

**Hüseyin Kaya**

**Student ID: 220303036**

**Date: 20 January 2026**

## Abstract

This report presents the implemented relational database system **Arel Global Market Place**, developed in **MySQL**. The design was implemented fully in SQL with referential integrity, validation rules, and automation using triggers.

The system models an Amazon-like marketplace including users, vendors, products/variants, orders, payments, shipments, and reviews. A key enhancement in the final version is **multi-currency support** via *Country–Currency mapping* and *FX rates*, allowing customer-facing prices to be stored in the customer's local currency while vendors keep base prices in their own currency.

## 1 Introduction

Modern e-commerce platforms require reliable databases to manage product catalogs, orders, payments, delivery, and customer feedback. This project implements a relational schema named *Arel Global Market Place* using **MySQL**.

The final version extends the initial design by adding: (1) **multi-currency transactions**, (2) **data validation constraints**, and (3) **business rules enforced via triggers**. It supports stock control, automatic order totals, verified reviews, and strict payment/refund logic.

# 2 Project Plan and Objectives

The project objectives are:

1. **Implement the Schema:** Create tables with keys, constraints, and indexes in MySQL.
2. **Preserve Integrity:** Enforce consistency using foreign keys and policies (cascade/restrict/set null).
3. **Enforce Rules:** Validate real-world constraints using triggers (age policy, stock control, verified reviews, etc.).
4. **Multi-Currency Support:** Derive customer currency from user country and apply FX conversion for pricing.
5. **Demonstration:** Insert sample data and verify correctness with SQL queries.

**Technology Used:** MySQL Database (InnoDB engine).

# 3 System Design and Scope

The database schema was designed with a clear separation of responsibilities in order to reflect real-world e-commerce workflows. Each logical layer focuses on a specific concern such as user localization, product management, transactional processing, logistics, or customer feedback. This modular structure improves data consistency, simplifies rule enforcement, and allows the system to be extended without major structural changes.

The implemented database contains **14 tables** organized into logical layers. In the final version, some relationships were redesigned to better support global commerce workflows (especially currency conversion and order/payment consistency).

- **1. User Layer:** *USERS* stores customer profile data with validation rules (e.g., age policy) and JSON-based phone numbers. The country is stored as *country_name* and mapped to currency using a separate table.

- **2. Localization & Currency Layer:** *COUNTRY_CURRENCY* maps each country to its official currency code. *FX_RATES* stores exchange rates between currency pairs. This layer enables the system to calculate customer prices based on the user's country currency.
- **3. Vendor & Catalog Layer:** *VENDORS*, *BRANDS*, and *CATEGORIES* represent sellers and the product taxonomy. Categories support hierarchy using a self-referencing foreign key (*parent_category_name*).
- **4. Product & Variant Layer:** *PRODUCTS* stores product-level information (SKU-based) and includes a semantic attribute *total_sold*. *PRODUCT_VARIANTS* stores sellable variants using a unique *stock_control_unit* key, with stock and pricing.
- **5. Order & Payment Layer:** *ORDERS* stores the order header with *order_code* and full shipping address snapshot fields. *ORDER_ITEMS* is a weak entity with composite primary key (*order_code*, *stock_control_unit*). Prices are stored both as vendor base price and customer currency price (*unit_price_customer*) derived using FX conversion. *PAYMENTS* is one-to-one with orders and supports strict refund rules.
- **6. Logistics & Review Layer:** *CARRIERS* and *SHIPMENTS* model delivery tracking. *REVIEWS* stores product feedback and is restricted to delivered purchases using trigger-based verification.

**Summary of Key Relationships (Final Version):**

1. 1 COUNTRY_CURRENCY → N USERS, N VENDORS (country_name FK)
2. 1 VENDORS → N PRODUCTS (vendor_name FK)
3. 1 CATEGORIES → N PRODUCTS (category_name FK), and category self-hierarchy
4. 1 PRODUCTS → N PRODUCT_VARIANTS (product_sku FK)
5. 1 ORDERS → N ORDER_ITEMS (order_code FK)
6. 1 ORDERS → 1 PAYMENTS (order_code UNIQUE FK)
7. 1 ORDERS → 1 SHIPMENTS (order_code UNIQUE FK)
8. 1 CARRIERS → N SHIPMENTS (carrier_name FK)
9. 1 USERS → N REVIEWS and 1 PRODUCTS → N REVIEWS (with purchase/delivery restrictions)

# 4 Results

The database was implemented and tested with sample data. Validation was ensured via constraints and trigger-based rules.

**Examples of enforced rules:**

- **Age policy:** users under 18 cannot be inserted/updated; under-18 users cannot place orders.
- **Phone validation:** *tel_no* must be valid JSON and numeric within limits.
- **Stock control:** order item insertion checks stock and decreases it automatically.
- **Auto totals:** order total is recalculated after inserting order items.
- **FX pricing:** customer currency is derived from user country; FX is applied to compute customer unit price.
- **Verified reviews:** reviews require purchase and delivery; duplicate reviews are blocked.

The following figure and table provide general context about global marketplace usage.

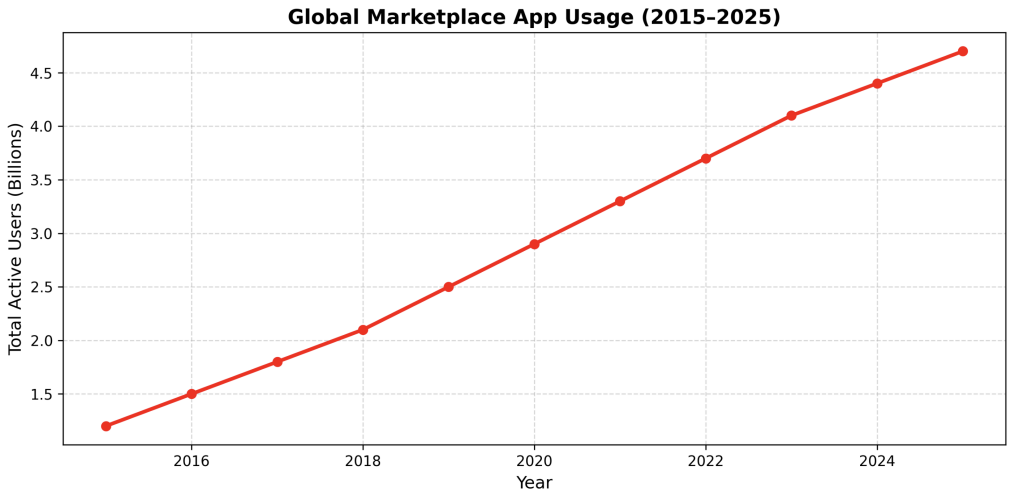


Figure 1: Global Marketplace App Usage (2015–2025)

| Country | Annual Revenue (Billion $) | Active Users (Million) |
|---|---|---|
| United States | 950 | 310 |
| China | 1250 | 720 |
| Germany | 310 | 95 |
| Turkey | 120 | 45 |
| India | 480 | 280 |
| Brazil | 260 | 120 |

Table 1: Example of Global Marketplace Data

# 5 Remarks and Recommendations

The final database moves from conceptual modeling to a working SQL implementation. The main improvements are multi-currency support, trigger-based business rules, and automation of stock and totals.

Future extensions can include a shopping cart mechanism for pre-order item management, discount and coupon support, order status history, product media tables, multilingual product descriptions, and auditing/logging, all of which can be integrated without affecting the existing core order and payment structure.

# References


1. Elmasri, R., & Navathe, S. B. (2016). *Fundamentals of Database Systems* (7th ed.). Pearson Education.

2. Silberschatz, A., Korth, H. F., & Sudarshan, S. (2019). *Database System Concepts* (7th ed.). McGraw-Hill Education.

3. MySQL. (n.d.). *MySQL 8.0 Reference Manual*. Retrieved from `https://dev.mysql.com/doc/`

4. W3Schools. (n.d.). *MySQL Tutorial*. Retrieved from `https://www.w3schools.com/mysql/`

5. GeeksforGeeks. (n.d.). *Enhanced ER Model in DBMS*. Retrieved from `https://www.geeksforgeeks.org`

6. dbdiagram.io. (n.d.). *Database Diagramming Tool*. Retrieved from `https://dbdiagram.io`